



Prepared with Synoptic

Executive briefing

Grounding on Boolarong Bank, Merrandi Channel, Kanooka Heads

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Contents

What happened	3
Why it happened.....	4
What we found.....	5
Contributing safety factors	5
Other factors that increased risk	5
Other findings.....	5
What is being done	7
Distribution	7

What happened

On 07/07/2026, MV Tarrakin Star was operating service TF18. She is a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers. TF18 was the last service of the day, an evening return from Muntari Island. The ferry departed at 17:35 with 38 passengers, a master, a deckhand and a kiosk attendant on board. Sunset had been at 17:02 and civil twilight had ended at 17:29, so the voyage was made in darkness. This had been true on every day the service had run since the winter timetable began on 01/06/2026, and this was the thirty-first such voyage.

Earlier that afternoon, the master had changed at the Kanooka Heads town wharf. The outgoing master told the relieving master that the chart plotter (the electronic chart display) had frozen twice that morning. Both times it had recovered by itself.

The ferry entered the Merrandi Channel from the north at 18:03:58. She held the charted centreline to within 3 m at 8.4 kt. Shortly afterwards, the deckhand was sent from the bridge to collect fares. At 18:07:06 the chart plotter froze again, and this time it did not recover. The vessel symbol stayed stationary on screen, and the display would not accept touch input. In the same second, the vessel's heading began to ease to starboard. At 18:08:31 the master began looking for the No. 4 beacon (a channel marker), and he sighted it 91 seconds later. Later checks found that mangroves screen the beacon from the channel between 0.84 nm and 0.34 nm north of it. By about 18:09 the ferry had crossed the western edge of the dredged channel on a held heading of 199 degrees. She was still making 8.0 kt, and no throttle movement was recorded.

At 18:11:48 the echo sounder shallow alarm sounded. The alarm had been set to warn at 0.9 m under the keel since the vessel was commissioned in 2019, and it gave 16 seconds' warning. Both throttles were put astern 4 seconds later. On trials, a crash stop from 8 kt takes 92 m, but only 58 m remained. At 18:12:04 the ferry struck Boolarong Bank at 6.8 kt. She was 74 m west of the centreline and 54 m outside the dredged channel. She stopped within 11 m and came to rest heeled 4 degrees to starboard on a falling tide.

The engines were stopped at 18:12:31, and the decision was made not to try to refloat. An urgency broadcast went to Port Control at 18:13:04, and Port Control called out the port workboat. All passengers were in lifejackets by about 18:22, although a locker behind the kiosk counter delayed the forward saloon. The workboat came alongside at 18:41, and the last passenger was ashore at 19:26.

There were no serious injuries. One passenger reported a grazed shin, and two were treated at the wharf for distress. The starboard hull was split, and a void flooded to approximately 260 mm. The starboard propeller, shaft and rudder were damaged, and the skeg was torn away. The ferry refloated at 00:20 and was withdrawn from service. The channel was closed to commercial traffic for 19 hours.

Why it happened

The ferry grounded because it drifted out of the dredged channel and was still travelling at 8 kt when the warning came, leaving too little room to stop. Its heading eased five degrees to starboard without any decision to turn, and it was virtually certain that this took the vessel out of the channel on the bank side.

The sequence began with the chart plotter freeze. It was extremely likely that the freeze contributed to the heading easing to starboard. It was also extremely likely that the freeze contributed to the transit continuing at 8 kt from that moment until the shallow alarm sounded. The plotter, moreover, had a history. The operator's defect procedure never required a fault that cleared itself to be recorded (Safety issue), and this likely contributed to the plotter freezing during the transit.

With the plotter lost, the No. 4 beacon was the obvious reference ahead. Mangroves, however, screened it from the channel between 0.84 nm and 0.34 nm north of it, and this likely contributed to the heading easing to starboard. Behind the screening lay an organisational cause. Clearing of the beacon's sightline had been deferred, unfunded, for three works rounds (Safety issue). It was very likely that this deferral contributed to the beacon being screened.

The speed was the second part of the problem. There was no night passage plan, and no hold point told the master what to do if the primary position fix was lost. It was very likely that this absence contributed to the ferry continuing at 8 kt. This gap, too, traced back to a decision made earlier in the year. The last service had been moved into darkness without the daylight-only risk assessment being reopened (Safety issue), and this likely contributed to there being no night passage plan or hold point.

Once the ferry was outside the channel, little margin remained. It was virtually certain that leaving the channel contributed to the shallow alarm giving only 16 seconds' warning, and extremely likely that the continued 8 kt did so as well. The alarm had been set at 0.9 m since 2019. It was extremely likely that this setting, and the short warning it produced, contributed to the grounding.

Two further matters were examined but could not be settled. The first concerns the unshielded boat ramp floodlights, which were aligned behind the No. 4 beacon when seen from the north. The alignment was real, and the masters' accounts of it were consistent and independent. Even so, the investigation could not establish either way whether the floodlights delayed the sighting in the seconds after the light cleared the islet, and so delayed the moment the held heading was noticed. It assessed this as about as likely as not. Whatever the answer, a lighting assessment of the channel is warranted.

The second concerns fatigue. Whether the master was fatigued was also about as likely as not. His duty and sleep made fatigue plausible, and he acknowledged being tired. The recorded performance, however, showed no sign of it. No detailed fatigue assessment was made. Every decision that mattered on the night was either explained by another factor or contradicted by the recorded data.

The investigation also identified other matters that increased risk but did not contribute to this grounding. These are reported with the findings.

What we found

A finding stated without a likelihood term was established as fact.

Contributing safety factors

- It was virtually certain that the last service was moved into darkness without the daylight-only risk assessment being reopened, and this likely contributed to the grounding. (Safety issue)
- It was virtually certain that there was no night passage plan and no hold point for loss of the primary fix (the main means of fixing the vessel's position), and it was very likely that this absence contributed to the grounding. (Safety issue)
- The operator's defect procedure never required a fault that cleared itself to be recorded, which was virtually certain, and this likely contributed to the grounding. (Safety issue)
- The chart plotter display froze during the channel transit and did not recover, which was virtually certain, and this very likely contributed to the grounding.
- With virtual certainty, the transit continued at eight knots from the plotter freeze until the shallow alarm, and this very likely contributed to the grounding.
- Clearing of the beacon's sightline was deferred, unfunded, for three works rounds, which was virtually certain, and this very likely contributed to the grounding. (Safety issue)
- The No. 4 beacon was screened by mangroves from the channel between 0.84 and 0.34 nm north of it, which was virtually certain, and this likely contributed to the grounding.
- The heading eased five degrees to starboard without a decision to turn, and this contributed to the grounding. Both were virtually certain.
- That the vessel left the dredged channel on the bank side was virtually certain, as was its contribution to the grounding.
- The shallow alarm at its 2019 setting of 0.9 m gave sixteen seconds' warning, which was virtually certain, and this very likely contributed to the grounding.

Other factors that increased risk

- The aids code set no notice or availability duty for beacons off declared channels, which was virtually certain. This did not contribute to the grounding, but it likely increased risk. (Safety issue)
- The beacon's range was halved by a solar conversion with no notice to channel users, which was virtually certain. Although this did not contribute to the grounding, it likely increased risk. (Safety issue)
- The darkness transit was rostered to close a split duty within 20 minutes of the spread limit, which was virtually certain. This did not contribute to the grounding, but it likely increased risk. (Safety issue)
- The forward saloon lifejacket locker was blocked by a counter fitted in 2019 without re-survey, which was virtually certain. While this did not contribute to the grounding, it very likely increased risk. (Safety issue)
- At the change of master, the plotter fault was passed on verbally and not entered in the defect book, which was virtually certain. This did not contribute to the grounding, but it likely increased risk.

Other findings

Fictional case study for training. Not a real investigation.

- The investigation examined whether the tide set down below its predicted height over Boolarong Bank and ruled it out, finding it very unlikely that any such set-down existed.
- A steering or propulsion failure before the vessel left the centreline was examined and ruled out, and it was found very unlikely that any such failure occurred.
- After the contact the engines were stopped and no attempt was made to refloat on a falling tide, which was virtually certain, and this very likely reduced risk.

What is being done

Last service moved into darkness without reopening the risk assessment. The General Manager Operations suspended evening Muntari services on 08/07/2026. [NOT ESTABLISHED: whether the suspension remains in place, and whether RRA-07 is to be reopened or a night passage plan written.]

No night passage plan or hold point for loss of the primary fix. The investigation has received no report of any action on this. [NOT ESTABLISHED: operator's response and its status.]

Defect procedure never requires a self-clearing fault to be recorded. Here too, no action has been reported in response. [NOT ESTABLISHED: operator's response to OPS-9 and its status.]

Clearing of the beacon's sightline deferred for three works rounds. Before this report was drafted, the investigation raised the No. 4 beacon with the Port Authority and the operator. Even so, no decision to fund or carry out the trimming has been reported. [NOT ESTABLISHED: status of WO-176 and the Authority's response.]

No notice or availability duty for beacons off declared channels. No action has been reported. Whatever the code requires, the Port Authority can close this gap through its own practice. [NOT ESTABLISHED: whether the Authority intends to do so.]

Beacon's range halved with no notice to channel users. This finding has been answered in part. Local notice LNM-2026-14 has been issued, and the Port Authority's aids to navigation register has been released. [NOT ESTABLISHED: whether the chart correction, pending on 07/07/2026, has been made, and whether the operator's passage plan still gives the beacon at 6 M.]

Darkness transit rostered close to the spread limit. No action has been reported. [NOT ESTABLISHED: operator's response on the winter roster, relief cover and fatigue diary review.]

Forward saloon lifejacket locker blocked by the kiosk counter. On 16/07/2026 the operator advised that the same counter is fitted to its two other 24 m vessels. The fleet-wide question now rests with the operator and the state regulator, and their answers are awaited. Meanwhile, when last reported, the locker on this vessel was still behind the counter. [NOT ESTABLISHED: corrective action on any of the three vessels.]

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